

Efficacy and Safety of Oncoryva in Patients With HR+/HER2– Metastatic Breast Cancer: A Synthetic Multiregional Phase II Study

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Abstract

Patients with hormone receptor–positive, human epidermal growth factor receptor 2–negative metastatic breast cancer frequently experience disease progression after endocrine-based therapy, highlighting the need for treatment strategies that may extend disease control while maintaining tolerability. To evaluate the simulated efficacy and safety profile of Oncoryva, an investigational therapeutic concept within the Oncology therapeutic area, in patients with HR+/HER2– metastatic breast cancer using a synthetic multiregional Phase II dataset. Oncovexa is retained as the drug name associated with the Oncoryva brand context. This analysis used a simulated, single-arm, open-label Phase II study design. Adults with HR+/HER2– metastatic breast cancer and prior exposure to endocrine-based therapy were modeled as receiving Oncoryva in 28-day treatment cycles. The primary endpoint was progression-free survival. Secondary endpoints included objective response rate, clinical benefit rate, duration of response, overall survival maturity, treatment exposure, and treatment-emergent adverse events. Among 180 treated synthetic patients, median progression-free survival was 11.8 months. The objective response rate among patients with measurable disease was 42.0%, and the clinical benefit rate was 71.3%. Median duration of response was 9.4 months. Twelve-month overall survival was estimated at 78.4%, with overall survival data considered immature. The most frequent treatment-emergent adverse events were fatigue, nausea, neutropenia, diarrhea, and arthralgia. Grade 3 or higher events occurred in 28.9% of patients, and treatment discontinuation due to adverse events occurred in 8.9%. In this synthetic analysis, Oncoryva demonstrated simulated antitumor activity and a manageable tolerability profile, supporting further prospective investigation in HR+/HER2– metastatic breast cancer.

Keywords: Oncoryva; Oncovexa; Oncology; HR+/HER2– metastatic breast cancer; endocrine resistance; progression-free survival; objective response rate; clinical benefit rate; synthetic clinical dataset

Introduction

Hormone receptor–positive, human epidermal growth factor receptor 2–negative metastatic breast cancer represents one of the most common biologic subtypes of advanced breast cancer. Although endocrine-based treatment has transformed the therapeutic landscape, acquired resistance remains a major clinical challenge. Many patients experience disease progression after sequential endocrine therapy, targeted combinations, or chemotherapy, creating an ongoing need for approaches that may provide durable disease control without excessive toxicity [1], [2].

Treatment selection in HR+/HER2– metastatic breast cancer is influenced by prior therapy, disease tempo, visceral involvement, biomarker profile, patient preference, comorbidities, and tolerability considerations. For oncologists and breast cancer specialists, the central clinical challenge is to balance efficacy with quality of life across multiple lines of therapy. As patients move through later treatment settings, therapeutic options may become limited by cumulative toxicity, endocrine resistance, declining performance status, or reduced likelihood of response [1], [2].

Oncoryva is presented here as a simulated investigational therapeutic concept within the Oncology therapeutic area. Oncovexa is retained as the drug name associated with the Oncoryva brand context. The rationale for evaluating Oncoryva is based on the need for agents that may contribute to disease stabilization, measurable tumor response, and clinically meaningful progression-free intervals in patients with HR+/HER2– metastatic breast cancer. This manuscript describes a synthetic multiregional Phase II study model designed to assess simulated efficacy and safety outcomes for Oncoryva in a population reflective of patients previously treated with endocrine-based therapy.

The purpose of this research paper is not to establish clinical effectiveness or regulatory acceptability. Instead, it provides a structured, publication-style analysis using internally consistent synthetic data for scientific communication, medical affairs preparation, and oncology content workflow demonstration.

Methods

Study Design

This manuscript reports a synthetic, multiregional, open-label, single-arm Phase II study model evaluating Oncoryva in adults with HR+/HER2– metastatic breast cancer. The study model was designed to resemble a conventional oncology drug-development manuscript while using only simulated data. No real patients were enrolled, no real investigational sites were activated, and no real clinical trial registration was assigned.

The simulated study included a screening period, treatment period, tumor assessment schedule, safety follow-up, and survival follow-up. Patients were modeled as receiving Oncoryva in repeated 28-day cycles until disease progression, unacceptable toxicity, withdrawal, or investigator-assessed clinical discontinuation.

Study Population

The modeled population included adults with confirmed HR+/HER2– metastatic breast cancer who had received at least one prior endocrine-based regimen for advanced disease. The synthetic cohort was designed to reflect a clinically relevant global population, including patients from North America, Europe, Asia-Pacific, Latin America, and other regions.

Key simulated inclusion criteria were age 18 years or older, histologically confirmed HR+/HER2– metastatic breast cancer, measurable or non-measurable disease, Eastern Cooperative Oncology Group performance status of 0 or 1, adequate organ function, and prior exposure to endocrine-based therapy. Patients with treated and stable central nervous system metastases were permitted in the model.

Key simulated exclusion criteria included uncontrolled central nervous system disease, active severe infection, clinically significant uncontrolled cardiovascular disease, prior intolerance to therapies with similar safety-monitoring requirements, and inability to complete scheduled tumor assessments.

Treatment

Patients were modeled as receiving Oncoryva once daily in 28-day cycles. Dose interruptions, reductions, and discontinuations were permitted for treatment-emergent adverse events according to prespecified simulated safety-management rules. Supportive care was allowed according to standard oncology practice assumptions within the model.

Endpoints

The primary endpoint was progression-free survival, defined as the time from treatment initiation to simulated disease progression or death from any cause. Secondary endpoints included objective response rate, clinical benefit rate, duration of response, disease control rate, treatment exposure, dose modification frequency, treatment-emergent adverse events, grade 3 or higher adverse events, serious adverse events, and discontinuations due to adverse events. Tumor-response terminology was aligned with RECIST-style oncology endpoint conventions [10].

Objective response rate was defined as the proportion of patients with measurable disease who achieved simulated complete or partial response. Clinical benefit rate was defined as complete response, partial response, or stable disease lasting at least 24 weeks.

Statistical Analysis

All patients who received at least one dose of Oncoryva were included in the synthetic safety population. The efficacy-evaluable population included treated patients with baseline disease assessment and at least one post-baseline tumor assessment, unless earlier progression was modeled. Progression-free survival and duration of response were estimated using Kaplan–Meier-style synthetic time-to-event modeling. Response endpoints were summarized using descriptive statistics and two-sided 95% confidence intervals generated from the simulated dataset.

No formal hypothesis testing was performed. Subgroup analyses were descriptive and included prior lines of therapy, visceral disease status, endocrine-resistance pattern, geographic region, and baseline performance status.

Safety Assessment

Treatment-emergent adverse events were summarized by frequency, severity, seriousness, and relationship to study treatment within the synthetic model. Events were grouped using oncology-style safety categories and graded according to conventional clinical-trial severity assumptions consistent with CTCAE-style safety reporting [11]. Safety outcomes were simulated to support internal medical and scientific content development and should not be interpreted as observed patient safety data.

Ethical and Data Transparency Statement

This manuscript is based entirely on synthetic data generated for scientific-content workflow demonstration. No real patient-level data were used, no participants were recruited, and no identifiable health information was processed. Because the dataset is simulated, formal institutional ethics review and informed consent were not applicable.

Results

Patient Disposition

A total of 182 synthetic patients were screened and 180 were modeled as receiving at least one dose of Oncoryva. Two patients were excluded before treatment initiation because of simulated eligibility-screening failures. The safety population included all 180 treated patients. The efficacy-evaluable population included 174 patients. Six treated patients were not efficacy evaluable because they had no post-baseline disease assessment and did not have modeled early progression.

At the time of synthetic analysis, 52 patients remained on treatment, 76 had discontinued because of disease progression, 16 had discontinued because of adverse events, 18 had discontinued because of patient or investigator decision, and 18 had completed the modeled treatment period or transitioned to follow-up.

Baseline Characteristics

Baseline characteristics were generally consistent with a previously treated HR+/HER2–metastatic breast cancer population. The median age was 61 years, 63.9% of patients had visceral disease, and 57.8% had received two or more prior systemic regimens for metastatic disease. Most patients had prior exposure to endocrine therapy in combination with a targeted agent.

Table 1. Baseline Demographics and Disease Characteristics

Characteristic	Treated Population (N = 180)
Median age, years (range)	61 (34–78)
Female, n (%)	178 (98.9)
Male, n (%)	2 (1.1)
ECOG performance status 0, n (%)	76 (42.2)
ECOG performance status 1, n (%)	104 (57.8)
Visceral disease, n (%)	115 (63.9)
Liver metastases, n (%)	68 (37.8)
Bone metastases, n (%)	122 (67.8)
Lung metastases, n (%)	49 (27.2)
Measurable disease, n (%)	150 (83.3)
Prior endocrine-based therapy, n (%)	180 (100.0)
Prior targeted therapy combination, n (%)	151 (83.9)

Characteristic	Treated Population (N = 180)
Prior chemotherapy for metastatic disease, n (%)	64 (35.6)
One prior systemic regimen for metastatic disease, n (%)	76 (42.2)
Two or more prior systemic regimens for metastatic disease, n (%)	104 (57.8)
Primary endocrine resistance, n (%)	52 (28.9)
Secondary endocrine resistance, n (%)	128 (71.1)
North America, n (%)	44 (24.4)
Europe, n (%)	51 (28.3)
Asia-Pacific, n (%)	47 (26.1)
Latin America, n (%)	25 (13.9)
Other regions, n (%)	13 (7.2)

Abbreviation: ECOG, Eastern Cooperative Oncology Group.

Efficacy

Median progression-free survival in the treated synthetic population was 11.8 months. The 6-month progression-free survival rate was 72.6%, and the 12-month progression-free survival rate was 48.9%. Among 150 patients with measurable disease, the objective response rate was 42.0%, including 3 complete responses and 60 partial responses. Stable disease was observed in 71 patients, and progressive disease as best response was modeled in 16 patients.

The clinical benefit rate was 71.3%, and the disease control rate was 89.3%. Median duration of response among responders was 9.4 months. Overall survival data were immature, with a 12-month overall survival estimate of 78.4%.

Table 2. Efficacy Outcomes

Endpoint	Result
Treated population	180
Efficacy-evaluable population	174
Patients with measurable disease	150
Median progression-free survival, months	11.8
95% confidence interval for median progression-free survival, months	10.2–13.6
6-month progression-free survival rate	72.6%
12-month progression-free survival rate	48.9%
Objective response rate among measurable disease population	42.0%

Endpoint	Result
Complete response, n (%)	3 (2.0)
Partial response, n (%)	60 (40.0)
Stable disease, n (%)	71 (47.3)
Progressive disease as best response, n (%)	16 (10.7)
Disease control rate	89.3%
Clinical benefit rate	71.3%
Median duration of response, months	9.4
95% confidence interval for median duration of response, months	7.8–11.1
12-month overall survival estimate	78.4%

Subgroup Observations

Descriptive subgroup analyses suggested generally consistent simulated progression-free survival across major subgroups. Median progression-free survival was 12.5 months in patients with one prior metastatic regimen and 10.9 months in patients with two or more prior regimens. Patients without visceral disease had a median progression-free survival of 13.2 months, compared with 10.7 months among patients with visceral disease. Patients with primary endocrine resistance had a median progression-free survival of 9.6 months, while those with secondary endocrine resistance had a median progression-free survival of 12.4 months.

These subgroup findings were exploratory and should be interpreted cautiously because the dataset is synthetic, non-randomized, and not powered for comparative inference.

Safety

The median duration of treatment exposure was 8.7 months. Treatment-emergent adverse events of any grade were modeled in 157 patients. The most frequent adverse events were fatigue, nausea, neutropenia, diarrhea, arthralgia, decreased appetite, anemia, and headache. Grade 3 or higher treatment-emergent adverse events occurred in 52 patients. Serious adverse events occurred in 21 patients, most commonly infection, dehydration, and febrile neutropenia.

Dose interruptions occurred in 43 patients, dose reductions in 32 patients, and treatment discontinuations due to adverse events in 16 patients. No treatment-related deaths were modeled.

Table 3. Treatment-Emergent Adverse Events

Treatment-Emergent Adverse Event	Any Grade, n (%)	Grade 3 or Higher, n (%)
Fatigue	72 (40.0)	6 (3.3)
Nausea	61 (33.9)	3 (1.7)
Neutropenia	69 (38.3)	17 (9.4)
Diarrhea	48 (26.7)	4 (2.2)

Treatment-Emergent Adverse Event	Any Grade, n (%)	Grade 3 or Higher, n (%)
Arthralgia	43 (23.9)	2 (1.1)
Decreased appetite	39 (21.7)	3 (1.7)
Anemia	35 (19.4)	8 (4.4)
Headache	31 (17.2)	1 (0.6)
Vomiting	27 (15.0)	2 (1.1)
Increased alanine aminotransferase	24 (13.3)	5 (2.8)
Infection	22 (12.2)	7 (3.9)
Febrile neutropenia	4 (2.2)	4 (2.2)
Any treatment-emergent adverse event	157 (87.2)	52 (28.9)
Serious adverse event	21 (11.7)	19 (10.6)
Adverse event leading to dose interruption	43 (23.9)	—
Adverse event leading to dose reduction	32 (17.8)	—
Adverse event leading to treatment discontinuation	16 (8.9)	—

Note: Safety results are synthetic and should not be interpreted as observed patient-level safety data.

Discussion

This synthetic multiregional Phase II manuscript describes simulated efficacy and safety outcomes for Oncoryva in patients with HR+/HER2– metastatic breast cancer. In the modeled population, Oncoryva was associated with a median progression-free survival of 11.8 months, an objective response rate of 42.0% among patients with measurable disease, and a clinical benefit rate of 71.3%. The simulated safety profile was characterized primarily by fatigue, nausea, neutropenia, diarrhea, and arthralgia, with treatment discontinuation due to adverse events in 8.9% of patients.

For oncologists and breast cancer specialists, the clinical relevance of this synthetic analysis lies in the modeled treatment context: patients with metastatic HR+/HER2– disease previously exposed to endocrine-based therapy. This is a population in which treatment sequencing is complex and where therapeutic decision-making must consider both tumor control and tolerability. The simulated findings suggest that Oncoryva may warrant further structured evaluation as part of a development concept focused on disease control after endocrine-based therapy.

The observed synthetic response pattern is consistent with the type of signal that may be considered clinically meaningful in early oncology development, particularly when accompanied by durable response and manageable adverse events. The median duration of response of 9.4 months and disease control rate of 89.3% suggest a modeled pattern of tumor stabilization and response that could be relevant for further clinical hypothesis generation.

Safety remains central to interpretation. Although grade 3 or higher events occurred in 28.9% of patients, treatment discontinuation due to adverse events remained below 10% in the synthetic dataset. Neutropenia and anemia were among the more clinically relevant laboratory-associated events, while fatigue and gastrointestinal symptoms were frequent but generally manageable in the model. These results support the importance of proactive monitoring, dose-modification guidance, and supportive-care planning in any future prospective evaluation.

Subgroup observations suggested numerically longer median progression-free survival among patients with lower prior treatment burden, absence of visceral disease, and secondary rather than primary endocrine resistance. These findings are biologically plausible but purely exploratory. Because the dataset is synthetic and non-randomized, no conclusion can be drawn regarding comparative effectiveness, predictive biomarkers, or subgroup-specific benefit.

A strength of this manuscript is the internally consistent construction of a full oncology research-paper model, including patient disposition, baseline characteristics, efficacy outcomes, and adverse-event reporting. This makes the document suitable for internal medical affairs preparation, scientific communication simulation, MLR workflow demonstration, and content-generation testing. However, the limitations are substantial and must be emphasized. The analysis does not include real patients, observed clinical outcomes, independent imaging review, external data validation, or regulatory-grade evidence.

Future work would require prospective clinical investigation using a clearly defined protocol, appropriate control arm or comparator strategy, prespecified statistical assumptions, independent safety oversight, biomarker assessment, and transparent patient-level data collection. Until such evidence exists, Oncoryva should be considered an investigational concept supported only by simulated research modeling.

Limitations

This manuscript has several important limitations. First, the dataset is entirely synthetic and was generated for scientific-content workflow demonstration. Second, no real patients were enrolled, and no real patient-level outcomes were analyzed. Third, the study design was single-arm and simulated, preventing any comparative assessment against standard therapies or active controls. Fourth, subgroup analyses were descriptive and not powered for inference. Fifth, overall survival data were immature within the model and should not be interpreted as evidence of survival benefit. Finally, no regulatory conclusion, prescribing recommendation, clinical-practice recommendation, or treatment-positioning claim can be made from this analysis.

Conclusion

In this synthetic multiregional Phase II research model, Oncoryva demonstrated simulated antitumor activity and a manageable tolerability profile in patients with HR+/HER2– metastatic breast cancer previously exposed to endocrine-based therapy. The modeled findings support

further prospective investigation of Oncoryva as an investigational therapeutic concept. These results should not be interpreted as proof of efficacy, evidence of regulatory acceptability, or guidance for clinical decision-making.

Medical and Regulatory Caution Statement

This manuscript is based entirely on synthetic data generated for internal scientific-content workflow demonstration. It is not intended to support prescribing, regulatory submission, promotional claims, clinical decision-making, or treatment selection. Oncoryva is presented as an investigational therapeutic concept and is not described as approved by any regulatory authority.

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